

Doctoral Program Oral Examination

**Trustworthy Hybrid Quantum–Classical AI for
Secure IoT Edge Systems: Secure Model
Deployment, Trusted Inference, and Zero-
Day Attack Detection**

by

Sajib Kumar Mitra

Announcement content

1. Previous work experience and job responsibilities
2. Research achievements (M.S. thesis, published papers, ongoing survey)
3. Motivation for pursuing higher education and doctoral research plan

Work experience

- **B.Sc. in Computer Science & Engineering**, University of Dhaka [2004 — 2010]
- **M.S. in Computer Science & Engineering**, University of Dhaka [2010 — 2012]
(Research: Reversible logic, quantum computing, low-power VLSI)
- **Software Engineer at Samsung Bangladesh R&D Center**, Dhaka [2011 — 2012]
(Worked in Commercial Development Team, SMP Mobile Platform)
- **Lecturer at Daffodil International University**, Dhaka, Bangladesh [2012 — 2014]
- **Joint Director (ICT) at Bangladesh Bank**, Dhaka, Bangladesh [2014 — Present]
(Large-scale secure ICT systems, cloud-native platforms, audit technology)
- **Researcher at University of Dhaka**, Dhaka, Bangladesh [Mar, 2026 — Present]
Co-investigator: Systematic survey on Quantum AI-powered IoT (57-paper corpus)

Responsibilities

- Secure ICT architecture, software engineering, and research leadership
- Conduct comprehensive ICT audits across multiple departments, evaluating system security, data integrity, operational compliance with regulatory standards.
- Implement web based data backup drive in PHP to safeguard critical information, employing encryption and regular backup to prevent data loss.

Research Achievements

Research spans quantum computing, reversible logic, hybrid IoT security, and embedded AI—with publications in IEEE/ACM venues.

	IEEE/ACM Journals	Conferences	Surveys	Other
Reversible & quantum logic design (M.S. era)	—	Multiple	—	—
Quantum ML & IoT security analytics	In progress	Multiple	1 (co-PI)	—
Hybrid edge intelligence & benchmarking	Planned	Planned	Foundation	Open artifacts
Professional ICT & secure systems	—	—	—	Industry scale
Total	Several	Several	1 major	Growing track

Ongoing survey — Quantum AI-powered IoT

(Survey manuscript)

- Smart-building IoT security gateway & post-discharge RPM IoMT case studies
- Hybrid Reference Architecture with policy-routed deployment paths α – γ

(PRISMA corpus $n=57$)

- Four-paradigm taxonomy: Conventional / AI / Quantum / QAI IoT
- Unified anomaly-detection benchmark $J(W)$ with latency & complexity metrics

Past presentations (M.S. thesis, published papers)

M.S. thesis — Reversible & quantum logic design

Thesis Paper: Design and Minimization of Reversible Programmable Logic Arrays and Its Realization using Pass Transistors (Published in 22th GLS VLSI, 2012)

Supervisor: Prof. Hafiz Md. Hasan Babu, Department of CSE, University of Dhaka

M.S. thesis Publication — Reversible & quantum logic design

S. K. Mitra, Lafifa Jamal, Mineo Kaneko, & Hafiz Md. Hasan Babu. 2012. "**An efficient approach for designing and minimizing reversible programmable logic arrays,**" In Proceedings of the great lakes' symposium on VLSI (GLS-VLSI'12). ACM, New York, NY, USA, 215-220, DOI: 10.1145/2206781.2206834.

Motivation for pursuing higher education

- Billions of IoT endpoints need trustworthy, energy-efficient security analytics.
- Hybrid Quantum AI at the edge is promising—but deployability remains unproven.
- **Things I want to cultivate in doctoral studies**
 - A. Rigorous research execution and reproducible benchmarking
 - B. Technical leadership bridging academic research and industry deployment
- **Goal:**
 - Bridge quantum ML research and deployable edge IoT security systems

PhD Research plan

Trustworthy Hybrid Quantum–Classical AI for Secure IoT
Edge Systems: Secure Model Deployment, Trusted
Inference, and Zero-Day Attack Detection

Background

Billions of IoT endpoints generate continuous telemetry under strict energy, latency, and trust constraints. Existing AI-based IoT security research focuses primarily on detection accuracy but rarely protects deployed AI models, inference pipelines, device identities, and cryptographic assets.

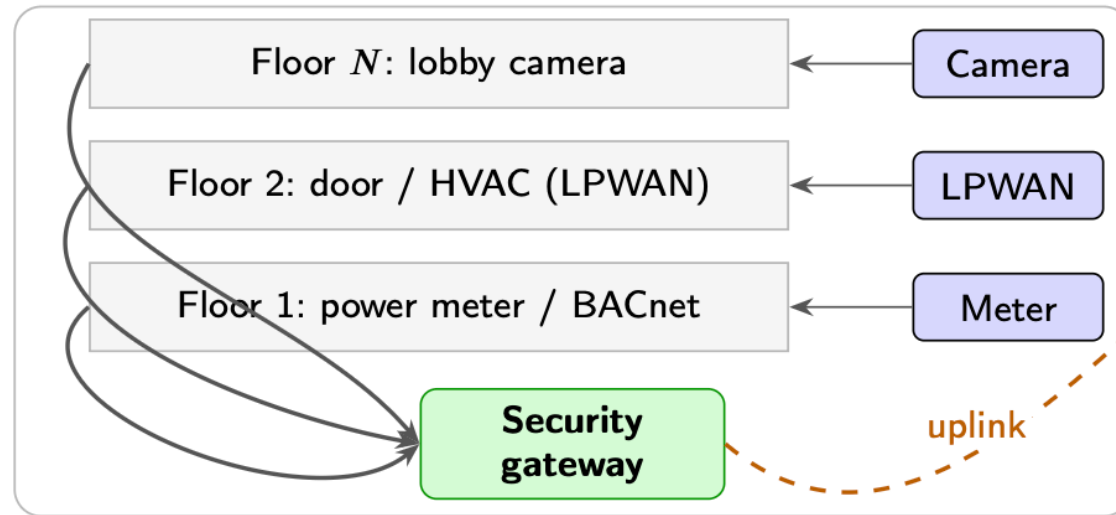


Quantum AI for IoT
Security

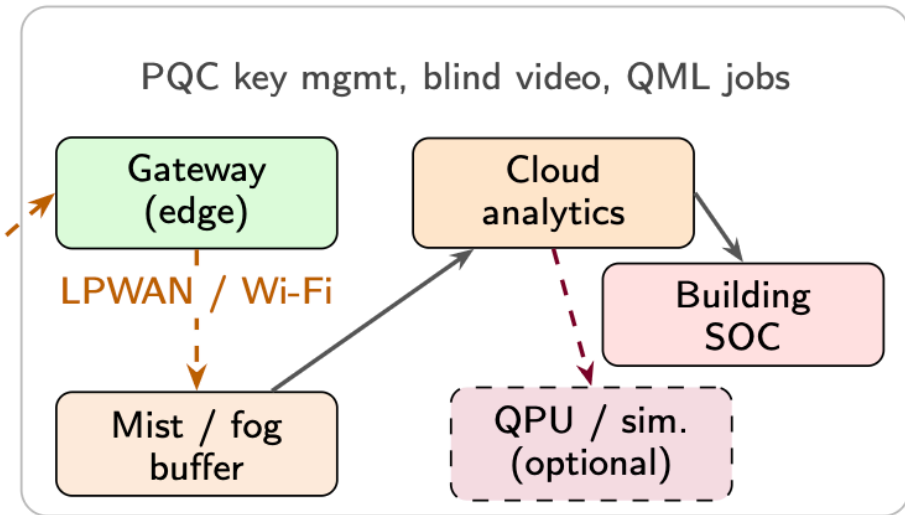
Practical IoT deployments require Trusted Execution Environments (TEE), Hardware Root of Trust, Secure Boot, Secure Key Management, and Remote Attestation.

Real Life Quantum AI-based IoT Application

A. Per-floor sensing (commercial tower)



B. Northbound operations



Research Questions

RQ1:

Can hybrid quantum-classical models improve zero-day and stealthy attack detection compared with lightweight classical AI under fixed edge constraints?

RQ2:

How can TEE, Root of Trust, Secure Boot, Remote Attestation, and Secure Key Management protect AI models and inference pipelines?

RQ3:

What security, latency, energy, and deployment trade-offs emerge when integrating Quantum AI with trusted edge-computing architectures?

Challenges and Attack Vectors

Attack Vector A: Zero-Day Malware and Novel Intrusions

Examples: Mirai variants, AI-generated malware, Unknown network intrusion patterns

Security Gap: Signature-based IDS cannot detect attacks that have never been seen before.

Attack Vector B: Lateral Movement in IoT Networks

Examples: Smart-building gateways, Industrial IoT devices, Smart healthcare devices

Security Gap: Traditional perimeter-based security cannot effectively detect internal propagation.

Attack Vector C: Resource-Constrained Evasion Attacks

Examples: CPU exhaustion attacks Memory exhaustion attacks Battery-drain attacks

Security Gap: Complex security analytics cannot be deployed on low-power edge devices.

Attack Vector D: Advanced Persistent and Stealthy Threats

Examples: Low-rate DDoS, Data exfiltration, Telemetry manipulation

Security Gap: Classical threshold-based monitoring often fails to distinguish these behaviors from normal activity.

Research Gap and Security pitfalls

A. Signature-Based Intrusion Detection Systems (Snort, Suricata, etc.)

Limitations:

- Require known attack signatures
- Ineffective against zero-day attacks
- Continuous manual updates are required

B. Classical Machine Learning-Based IDS (Random Forest, SVM, CNN, LSTM)

Limitations:

- High computational cost
- Large memory footprint
- Difficult deployment on edge gateways

C. Cloud-Based Security Analytics

Limitations:

- High communication latency
- Privacy concerns
- Dependence on continuous connectivity

D. Others

Limitations:

- Lack of end-to-end trustworthy architectures integrating anomaly detection
- Trusted execution and Secure AI deployment
- Device authentication and Lifecycle protection.

Related research

Prior QML for IoT anomaly detection

→ Competitive accuracy but substantial simulator-era overhead

S. A. Chowdhury, S. F. Karim, M. S. I. Bhuiyan, A. A. Shuvro, and M. S. Hossen. Quantum machine learning for anomaly detection in iot devices. In Proc. Int. Conf. AI-ML-Systems (AIMLSystems), pages 406–411, 2025. doi: 10.1109/aimlsystems67835.2025.11331191.

Projected quantum kernels for IoT

→ Cloud-centric; limited edge deployability evidence

F. D’Amore, L. Mariani, C. Mastroianni, J. Settino, and A. Vinci. Projected quantum kernel for iot data analysis. In Proc. IEEE Conf. Pervasive Intell. Comput. (PCom), 2024. doi: 10.1109/picom64201.2024.00032.

Embedded QML feasibility studies

→ Hybrid architectures proposed; empirical edge validation still sparse

S. Dey and S. M. Raza. Embedded quantum machine learning in embedded systems: Feasibility, hybrid architectures, and quantum co-processors, 2026.

Research Gap

Despite rapid QML progress, four barriers block practical edge adoption:

- 1. Cloud-centric training assumptions:** Variational quantum circuits (VQCs) are sized for cloud QPUs, not microcontroller or FPGA budgets.
- 2. Missing metric alignment:** Studies rarely report accuracy jointly with circuit depth, gate count, inference latency, and energy per decision on comparable IoT workloads.
- 3. Weak deployment path:** Few prototypes demonstrate train-on-cloud/simulator, infer-on-edge pipelines with open datasets and reproducible baselines.
- 4. Over-broad claims:** Theoretical quantum advantage statements seldom specify edge power caps, target false-alarm rates, or classical strong baselines.

Core Security Problem

Targeted Security Threats:

- Zero-day malware attacks
- Lateral movement attacks
- Firmware tampering
- AI model extraction
- AI model poisoning
- Unauthorized device enrollment
- Supply-chain attacks
- Advanced persistent threats

Research Goal:

- IoT infrastructure
- AI assets deployed on edge devices

Research Approach

Properties of Trustworthy Hybrid Quantum-Classical Security Framework

- Edge Data Collection
- Hybrid Quantum-Classical Threat Detection
- Trusted AI Deployment
- Secure Inference Execution
- Remote Attestation & Device Verification
- Continuous Security Monitoring

Experimental Platform

More realistic and reproducible experimental platforms to evaluate work

- **AI Layer**

- Qiskit
- PennyLane
- VQC
- Quantum Kernels

- **Trusted Computing Layer**

- ARM TrustZone
- OP-TEE

- **Edge Platform**

- Raspberry Pi
- ARM Gateway

- **Acceleration Platform**

- Xilinx Zynq FPGA

- **Quantum Backend**

- IBM Quantum

Trustworthy AI Deployment Architecture

The research beyond anomaly detection to investigate a secure AI deployment architecture incorporating:

- Trusted Execution Environments (TEE)
- Hardware Root of Trust
- Secure Boot
- Secure Key Management
- Remote Attestation
- Trusted AI Model Deployment and Inference
- Protect AI models from:
 - Model Theft
 - Model Tampering
 - Unauthorized inference
 - Firewall compromise

Research Workflow

Step 1: Edge-Based Feature Collection

Collect Data from multiple sources:

- Network flows
- Device telemetry
- System logs
- Gateway traffic

Step 2: Hybrid Quantum-Classical Learning

- Qiskit / PennyLane on simulators or cloud QPUs;
- hyper-parameter search under qubit-depth constraints;
- optional post-quantum security context for threat models (performs Complex pattern learning, Anomaly classification, Zero-day attack detection).

Research Workflow

Step 3: Edge Deployment

- Compressed model export to FPGA bitstream or optimized C++/TinyML runtime on Raspberry Pi / industrial gateway;
- local anomaly scoring on streaming features.

Step 4: Evaluation phase

- Unified metrics aligned with the candidate's survey benchmark table (complexity class, wall-clock latency, energy, F1/AUC).
- Datasets (KDDCup99, anoML-IoT, TWTDUS, IoT telemetry datasets)

Evaluation

Trusted Security Metrics

- Secure Boot Success Rate
- Attestation Success Rate
- Model Tampering Detection Rate
- Model Extraction Resistance
- Key Management Overhead

AI Metrics

- Accuracy
- Precision
- Recall
- F1
- AUC

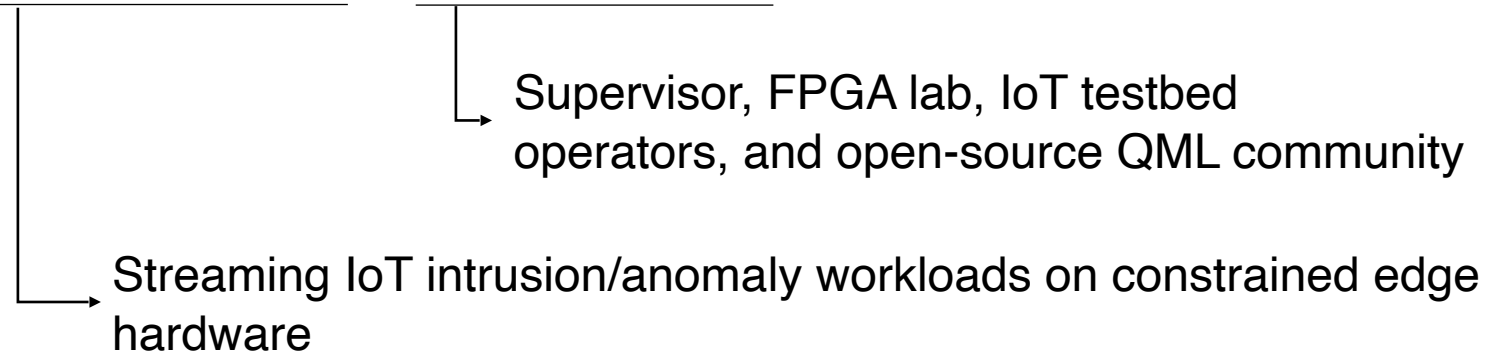
Platform Metrics

- Latency
- Energy
- Memory
- FPGA Resource

Research steps

A. Benchmark protocol & requirements (Year 1)

- PRISMA-inspired literature update; dataset selection (Scopus, WoS, PubMed, etc.)
- Reproducible harness: accuracy, latency, algorithmic complexity, energy proxy
- Classical edge baselines (lightweight CNN / autoencoder)



- Define stakeholders: sensor owners, edge operators, cloud trainers, end users.
- Clarify trust & security requirements for deployable hybrid edge QML.

Research steps

B. Hybrid model development & hardware integration (Years 2–3)

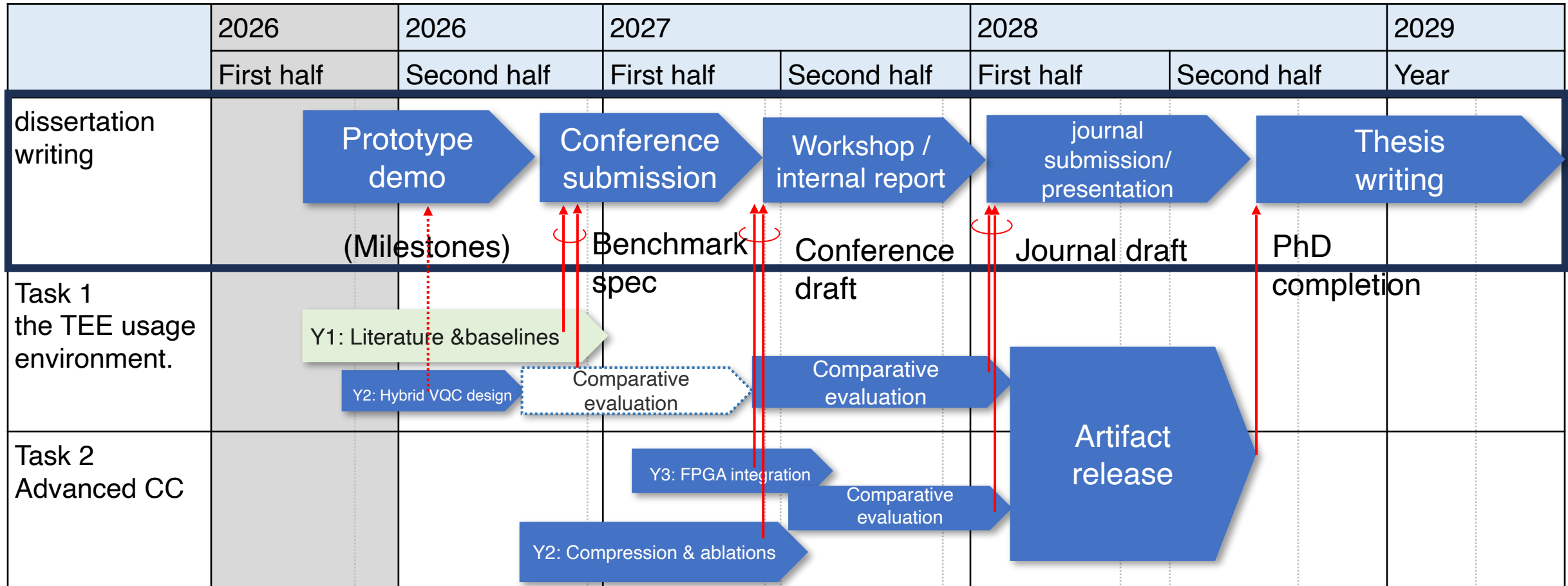
- VQC / quantum-kernel design with noise-aware training and compression ablations
- FPGA/embedded integration (Vivado/Vitis); power measurement via USB meter or PMIC
- Comparative evaluation vs. classical TinyML and cloud-hybrid QML under fixed budgets

C. Trusted AI Deployment Architecture

- Safety evaluation: threat analysis; negative results published when hybrid QML is not justified.
- Optional Year 2: reversible-logic data-path block with switching-activity measurement.
- Focus on secure Development and Trustworthy execution of proposed Framework

Announcement plan (48 months)

Research proceeds iteratively across benchmarking, hybrid model design, edge prototyping, and thesis synthesis. Results target IEEE IoT Journal, FGCS, and ACSAC, RAID, ESORICS, DIMVA, AsiaCCS, or other comparable security conferences.



▲ Start

Completion ▲

In conclusion (Aspirations)

Long-Term Vision:

- Develop trustworthy AI-enabled IoT security systems
- Integrate Quantum AI with secure deployment architectures
- Bridge academic Quantum ML research and industry-scale secure systems
- Contribute to practical, deployable, and resilient IoT infrastructures